Integrative Enrichment of Protein Functional Annotations Using Large Language Models

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1 Motivation

Protein functional annotation is a major challenge in bioinformatics, as many sequenced proteins still lack a complete and reliable functional characterization. This is a problem that has become even more evident with the rapid growth of biological data, especially data generated by high-throughput technologies, which has increased both the volume and the complexity of the information that needs to be interpreted. Although several computational tools have been developed to support protein function prediction, their results are often distributed across different outputs and presented in heterogeneous formats. In practice, this means that users often need to install and use multiple tools in order to obtain complementary information. This gives rise to a new challenge, namely the need to develop a solution capable of organizing, integrating and interpreting these results in a consistent way as such a solution would be highly useful to support research in protein analysis. The present project is motivated by this need. The objective is to improve the consistency and interpretability of protein annotation results through the integration and harmonization of previously generated outputs, enriched with complementary information from biological databases, with processing supported by large language models.

2 State of the Art

To understand the current context of bioinformatics it is important to consider the impact of the rapid growth of biological data and its implications for data analysis. The significant increase in data volume, combined with the diversity of available approaches and tools has contributed to the difficulty of obtaining consistent results, particularly in the field of protein functional annotation. This scenario highlights the need to understand the main existing approaches and their limitations.

2.1 Growth of Biological Data

Over the last years, bioinformatics has been increasingly influenced by the rapid growth of biological data. This growth is not only related to the amount of information being generated, but also to the complexity and heterogeneity of

biological data [1]. The ability to acquire, process, and analyse large biological datasets is now central to extracting useful knowledge from the available information [2]. Next-generation sequencing (NGS) has revolutionized genomics by enabling the parallel sequencing of millions of DNA fragments, with high processing capacity and improved cost-effectiveness [3]. It is characterized by the use of shorter reads and relies on massively parallel sequencing to obtain an accurate consensus sequence [4]. Advances in high-throughput DNA sequencing and bioinformatics have made it possible to systematically explore the genetic diversity present in complex microbial communities directly from environmental samples, without relying on prior isolation of individual microorganisms [5]. The growing amount of biological data has fuelled the development of omics approaches, transforming the way biology is studied and understood. These approaches analyse biological information across multiple layers at a large scale, enabling a more comprehensive understanding of organisms and their interactions [6]. Thus, NGS has facilitated the generation of complete genomic sequencing data, shifting the focus of research towards genomic annotation. When combined with metagenomics, it has significantly advanced research. However, the extraction of useful information from these data remains limited by the lack of functional information[7].

2.2 Protein Annotation Challenges

The exponential growth of biological data has resulted in a large number of protein sequences. Although the number of sequences has increased substantially, functional protein annotation has not kept pace with this growth. Approximately 50% of the proteins predicted across more than 500,000 sequenced genomes possess reliable functional annotations, leaving a major gap in the ability to interpret these data effectively [8]. Functional annotation of proteins is crucial for understanding biological mechanisms, as without accurate functional assignment, it is difficult to understand how proteins interact and the lack of rigorous annotation hinders the use of these sequences for practical applications, such as the development of new therapeutic treatments. [9, 10] Despite the limitations, computational methods for protein function prediction have advanced considerably. Traditional approaches, including sequence similarity-based methods, structural modeling, and phylogenetic analysis, have long been used to infer protein function [11]. More recently, machine and deep learning techniques have been widely applied to predict functions based on existing data, and have shown increasing effectiveness, particularly when integrated with experimental evidence [12].

2.3 Computational Tools for Protein Function Prediction

Computational methods have been developed across different methodological classes to support protein function prediction. The integration of traditional computational methods and machine learning techniques has significantly advanced the understanding of protein function and protein engineering. Since proteins are the basis of virtually all biological processes, accurate predictions of their

functions, whether based on sequence or structural data, are essential for their characterization [10, 13]. Current computational approaches for protein function prediction can be broadly grouped into three main categories: homology-based, structure-based, and machine-learning-based methods. Although these approaches differ in the type of evidence they rely on, they all aim to support the functional characterization of proteins that remain unannotated or only partially characterized. Recent studies have shown that protein function prediction benefits from integrating multiple types of evidence rather than relying on a single type alone [11, 14]. Within these categories, several open-source tools are available and generate different types of annotation outputs. In this context, homology-based approaches typically produce results such as functional descriptions, conserved domain assignments, Gene Ontology (GO) terms, Enzyme Commission (EC) numbers, taxonomic information, and pathway-related annotations. Representative examples of open-source tools in this group include reCOGnizer and UPIMAPI [15], as well as eggNOG-mapper [16]. Structure-based approaches provide a different type of information, often derived from predicted three-dimensional structures, structural comparisons, similarity searches, or structure-informed functional inference. Open-source tools such as ColabFold [17], Foldseek [18] and DeepFRI [19] are examples of methods that generate this type of output. Machine-learning-based approaches have also become increasingly relevant, particularly for proteins that are difficult to characterize using sequence similarity alone. In general, these methods typically generate predicted functional annotations such as GO terms or EC numbers from protein-related data. Examples include DeepGO-based approaches [20] as well as CLEAN [21]. Overall, these tools generate heterogeneous annotation outputs rather than a single unified type. This heterogeneity reflects the fact that protein function can be inferred from multiple biological perspectives, with each approach contributing a distinct layer of evidence. Consequently, integrating multiple biological modalities for protein function prediction is essential, as complementary evidence can improve the robustness and interpretability of functional predictions [11, 14]. However, this diversity also introduces challenges, as outputs from different tools must be systematically organized, and harmonized to enable consistent interpretation.

2.4 Biological Databases for Protein Annotation

The integration of multiple information sources is essential for improving protein function prediction [22]. In this context, biological databases play a central role in modern science, despite the challenges associated with the massive growth of biological data and the need to make such data accessible to the scientific community [23]. However, the integration of large amounts of heterogeneous information from different databases and formats remains a major challenge [24]. One of the most relevant databases for protein annotation is UniProt. This resource aims to provide a comprehensive, high-quality, and freely accessible collection of protein sequences annotated with functional information [25]. It integrates metadata from more than 180 interconnected databases, assigns stable

protein identifiers, and includes an ID Mapping tool that facilitates data search, retrieval, and integration in computational workflows [24]. Another highly relevant database is STRING. Protein function is often mediated through interactions, making protein–protein association data essential for understanding cellular processes. STRING evaluates and integrates information on protein-protein associations derived from experimental assays, computational predictions, and prior knowledge. It also provides clustering and enrichment analysis tools that support the functional interpretation of protein sets [26]. The complementarity of these databases is particularly relevant, as protein function prediction increasingly benefits from the integration of multiple types of evidence [11]. In this way, combining information from different databases, such as UniProt and STRING, enables a more complete and consistent interpretation of protein function. When integrated with heterogeneous outputs from computational prediction tools, these resources further contribute to improving and contextualizing the available functional information.

2.5 APIs for Automated Data Retrieval and Integration

Given the large volume of results generated, manually consulting databases to add relevant information becomes inefficient. Once the results have been normalized, database APIs can be used to complement and enrich the information initially obtained from the tool outputs [24]. The use of APIs enables automated queries and the retrieval of structured information in an efficient and standardized manner, facilitating the integration of large and heterogeneous bioinformatics datasets and contributing to more comprehensive protein annotations [27]. The UniProt APIs are open resources based on the UniProt.org website and allow for flexible programmatic interaction with protein knowledge data. They provide access to resources such as UniProtKB, UniRef, UniParc, Proteomes, GeneCentric, ARBA, UniRule, and the ID Mapping tool. They also allow for structured queries, the selection of specific fields, and the downloading of results in multiple formats. Since several UniProt services are connected through shared identifiers and data types, these APIs are particularly useful for identifier mapping and for gathering dispersed protein-related information in a more organized way [24]. They are especially valuable for retrieving structured functional annotations and stable identifiers that can help enrich the information obtained from the initial tool outputs. The STRING API allows programmatic access to STRING data and is one of the official access methods, being especially useful for automated workflows. The STRING ecosystem uses these web-service APIs for tasks such as identifier mapping, network import, and functional enrichment retrieval, allowing for the automatic and reusable extraction of relevant functional association information [26, 28]. This functionality is particularly useful for complementing initial annotation results with information on protein associations, interaction networks, and a broader functional context, thereby adding an additional layer of evidence for protein interpretation. After information is automatically collected through the different APIs, the resulting outputs remain heterogeneous. Therefore automated retrieval alone is not sufficient to ensure a

fully coherent interpretation of protein annotations. Instead, the main role of this step is to collect and enrich the information that will subsequently need to be organized and harmonized.

2.6 Large Language Models for Data Harmonization

Large Language Models (LLMs) are trained on large volumes of text and are known for their ability to generate and process natural language [29]. Their relevance has grown in recent years, as they can support tasks such as information summarization, knowledge extraction, and content organization [30]. LLMs are particularly important when information collected from different sources is expressed through distinct labels, descriptions, or semantics and therefore needs to be aligned coherently. In this sense, they can help harmonize heterogeneous biological metadata by transforming inconsistent entries into more standardized representations [31]. In protein functional annotation, this capability becomes relevant because different tools may produce GO terms, EC numbers, functional descriptions, conserved domains, or structural evidence that are not always presented uniformly. Although these results may be complementary, they can also be redundant, partially overlapping, or semantically difficult to compare directly. After the initial results have been normalized and enriched with information retrieved from external databases, LLMs can be used to address this residual heterogeneity [32, 31]. In this way, LLMs appear here as a final semantic layer of harmonization. They support the integration and reorganization of the annotations already obtained, contributing to the production of a more consistent and interpretable final functional description for the proteins under study.

3 Objectives

The main objective of this work is to develop a process for the integration and harmonization of protein functional annotation results obtained from outputs generated by different open-source tools from a previous study. This process aims to improve the consistency and interpretability of the available annotations and to support the production of a more informative final protein annotation.

4 Proposed Workflow

The present work will be developed as a bioinformatics workflow for the integration, enrichment, and harmonization of protein functional annotation results obtained from outputs generated by different open-source functional prediction tools applied in a previous study. As illustrated in Figure 1, the proposed workflow is organized into sequential stages, starting with the selection of input files and ending with the production of an integrated and harmonized final output. The output files resulting from these tools will be selected and imported into a common computational environment. These files, originally produced by different protein functional annotation approaches, will be converted into tabular

structures in Python in order to facilitate subsequent processing. A normalization step will then be applied with the aim of standardizing identifiers, column names, and annotation fields, creating an intermediate dataset with a consistent structure across the different sources. After normalization, an identifier mapping stage will be carried out to reconcile different designations that refer to the same protein. This step will include identifier cleaning, format adjustment, and conversion into a more stable reference, allowing annotations from different files to be associated at the protein level. Once the data have been normalized and mapped, the workflow will proceed to the enrichment stage through automated queries to the UniProt and STRING APIs. This step will allow the addition of structured annotation fields, including protein and gene names, functional descriptions, Gene Ontology terms, Enzyme Commission numbers, and information on protein associations. The retrieved data will then be integrated with the normalized results in order to consolidate the available evidence into a unified protein-centered representation. In the final stage, the integrated dataset will be submitted to an LLM-based harmonization process. The objective of this phase will be to reorganize the collected annotations, reduce redundancies, align semantically similar descriptions, and generate a clearer and more coherent functional summary for each protein. Thus, the final output of the workflow will consist of a protein-level table containing the normalized and enriched annotations, together with a harmonized textual synthesis to support the interpretation of the final results.

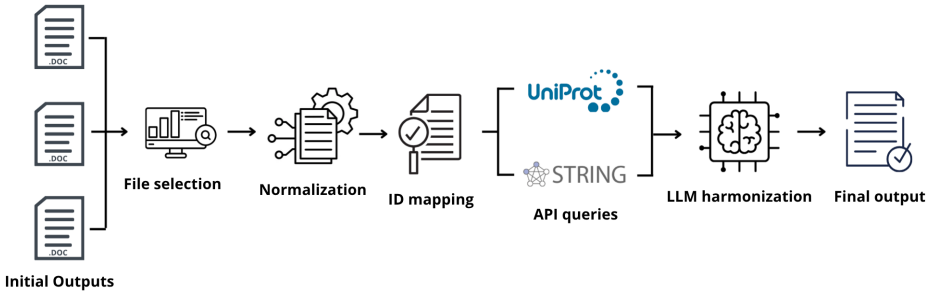

Fig. 1. Practical workflow representation of the proposed integration pipeline. Initial output files are selected, normalized, mapped, enriched through UniProt and STRING API queries, and finally harmonized using LLMs to generate the final output.

References

1. Gabriel Dall'alba, Pedro Lenz Casa, Fernanda Pessi De Abreu, Daniel Luis Notari, and Scheila De Avila E Silva. A survey of biological data in a big data perspective. *Big Data*, 10(4):279–297, 2022.

2. Prakash Kumar, Ranjit Kumar Paul, Himadri Shekhar Roy, Md Yeasin, Ajit, and Amrit Kumar Paul. Big data analysis in computational biology and bioinformatics. *Methods in Molecular Biology*, 2719:181–197, 2024.
3. Heena Satam, Kandarp Joshi, Upasana Mangrolia, Sanober Waghoo, Gulnaz Zaidi, Shravani Rawool, Ritesh P. Thakare, Shahid Banday, Alok K. Mishra, Gautam Das, and Sunil K. Malonia. Next-Generation Sequencing technology: Current trends and advancements. *Biology*, 12(7):997, 2023.
4. Barton E. Slatko, Andrew F. Gardner, and Frederick M. Ausubel. Overview of next generation sequencing technologies. *Current protocols in molecular biology*, 122(1):e59, 2018.
5. Thomas J. Sharpton. An introduction to the analysis of shotgun metagenomic data. *Frontiers in Plant Science*, 5(JUN):209, 2014.
6. Mingon Kang, Euiseong Ko, and Tesfaye B. Mersha. A roadmap for multi-omics data integration using deep learning. *Briefings in Bioinformatics*, 23(1):bbab454, 2022.
7. Girum Fitihamlak Ejigu and Jaehee Jung. Review on the computational genome annotation of sequences obtained by Next-Generation Sequencing. *Biology 2020, Vol. 9, Page 295*, 9(9):295, 2020.
8. Valérie De Crécy-Lagard, Rocio Amorin De Hegedus, Cecilia Arighi, Jill Babor, Alex Bateman, Ian Blaby, Crysten Blaby-Haas, Alan J. Bridge, Stephen K. Burley, Stacey Cleveland, Lucy J. Colwell, Ana Conesa, Christian Dallago, Antoine Danchin, Anita De Waard, Adam Deutschbauer, Raquel Dias, Yousong Ding, Gang Fang, Iddo Friedberg, John Gerlt, Joshua Goldford, Mark Gorelik, Benjamin M. Gyori, Christopher Henry, Geoffrey Hutinet, Marshall Jaroch, Peter D. Karp, Liudmyla Kondratova, Zhiyong Lu, Aron Marchler-Bauer, Maria Jesus Martin, Claire McWhite, Gaurav D. Moghe, Paul Monaghan, Anne Morgat, Christopher J. Mungall, Darren A. Natale, William C. Nelson, Seán O'Donoghue, Christine Orengo, Katherine H. O'Toole, Predrag Radivojac, Colbie Reed, Richard J. Roberts, Dmitri Rodionov, Irina A. Rodionova, Jeffrey D. Rudolf, Lana Saleh, Gloria Sheynkman, Francoise Thibaud-Nissen, Paul D. Thomas, Peter Uetz, David Vallenet, Erica Watson Carter, Peter R. Weigele, Valerie Wood, Elisha M. Wood-Charlson, and Jin Xu. A roadmap for the functional annotation of protein families: a community perspective. *Database*, 2022:baac062, 2022.
9. Jiajun Hong, Yongchao Luo, Yang Zhang, Junbiao Ying, Weiwei Xue, Tian Xie, Lin Tao, and Feng Zhu. Protein functional annotation of simultaneously improved stability, accuracy and false discovery rate achieved by a sequence-based deep learning. *Briefings in Bioinformatics*, 21(4):1437–1447, 2019.
10. Tian Ci Yan, Zi Xuan Yue, Hong Quan Xu, Yu Hong Liu, Yan Feng Hong, Gong Xing Chen, Lin Tao, and Tian Xie. A systematic review of state-of-the-art strategies for machine learning-based protein function prediction. *Computers in Biology and Medicine*, 154:106446, 2023.
11. Baohui Lin, Xiaoling Luo, Yumeng Liu, and Xiaopeng Jin. A comprehensive review and comparison of existing computational methods for protein function prediction. *Briefings in Bioinformatics*, 25(4):bbae289, 2024.
12. Ekaterina V. Ilgisonis, Pavel V. Pogodin, Olga I. Kiseleva, Svetlana N. Tarbeeva, and Elena A. Ponomarenko. Evolution of protein functional annotation: Text mining study. *Journal of Personalized Medicine*, 12(3):479, 2022.
13. Yaan J. Jang. Editorial: Computational protein function prediction based on sequence and/or structural data. *Frontiers in Bioinformatics*, 5:1705252, 2025.

14. Wenjian Ma, Xiangpeng Bi, Huasen Jiang, Zhiqiang Wei, and Shugang Zhang. Annotating protein functions via fusing multiple biological modalities. *Communications Biology*, 7(1):1705, 2024.
15. João C. Sequeira, Miguel Rocha, M. Madalena Alves, and Andreia F. Salvador. UpimAPI, reCOGnizer and KEGGCharter: Bioinformatics tools for functional annotation and visualization of (meta)-omics datasets. *Computational and Structural Biotechnology Journal*, 20:1798–1810, 2022.
16. Carlos P. Cantalapiedra, Ana Hernandez-Plaza, Ivica Letunic, Peer Bork, and Jaime Huerta-Cepas. eggnog-mapper v2: Functional annotation, orthology assignments, and domain prediction at the metagenomic scale. *Molecular Biology and Evolution*, 38:5825–5829, 12 2021.
17. Milot Mirdita, Konstantin Schütze, Yoshitaka Moriwaki, Lim Heo, Sergey Ovchinnikov, and Martin Steinegger. ColabFold: making protein folding accessible to all. *Nature Methods 2022 19:6*, 19(6):679–682, 2022.
18. Michel van Kempen, Stephanie S. Kim, Charlotte Tumescheit, Milot Mirdita, Jeongjae Lee, Cameron L.M. Gilchrist, Johannes Söding, and Martin Steinegger. Fast and accurate protein structure search with Foldseek. *Nature Biotechnology*, 42(2):243–246, 2024.
19. Vladimir Gligorijević, P. Douglas Renfrew, Tomasz Kosciolek, Julia Koehler Leman, Daniel Berenberg, Tommi Vatanen, Chris Chandler, Bryn C. Taylor, Ian M. Fisk, Hera Vlamakis, Ramnik J. Xavier, Rob Knight, Kyunghyun Cho, and Richard Bonneau. Structure-based protein function prediction using graph convolutional networks. *Nature Communications*, 12(1):3168, 2021.
20. Maxat Kulmanov, Mohammed Asif Khan, and Robert Hoehndorf. DeepGO: predicting protein functions from sequence and interactions using a deep ontology-aware classifier. *Bioinformatics*, 34(4):660–668, 2018.
21. Tianhao Yu, Haiyang Cui, Jianan Canal Li, Yunan Luo, Guangde Jiang, and Huimin Zhao. Enzyme function prediction using contrastive learning. *Science*, 379(6639):1358–1363, 2023.
22. Maxat Kulmanov and Robert Hoehndorf. Computational prediction of protein functional annotations. *Methods in Molecular Biology*, 2947:3–28, 2025.
23. Mikołaj Danielewski, Marlena Szalata, Jan Krzysztof Nowak, Jarosław Walkowiak, Ryszard Słomski, and Karolina Wielgus. History of biological databases, their importance, and existence in modern scientific and policy context. *Genes*, 16(1):100, 2025.
24. Shadab Ahmad, Leonardo Jose Da Costa Gonzales, Emily H. Bowler-Barnett, Daniel L. Rice, Minjoon Kim, Supun Wijerathne, Aurélien Luciani, Swaathi Kandasaamy, Jie Luo, Xavier Watkins, Edd Turner, Maria J. Martin, Alex Bateman, Maria Jesus Martin, Sandra Orchard, Michele Magrane, Shiqi Ye, Aduragbemi Adesina, Shadab Ahmad, Emily H. Bowler-Barnett, David Carpentier, Paul Denny, Jun Fan, Leonardo Jose Da Costa Gonzales, Abdulrahman Hussein, Alexandr Ignatchenko, Giuseppe Insana, Rizwan Ishtiaq, Vishal Joshi, Dushyanth Jyothi, Swaathi Kandasaamy, Antonia Lock, Aurelien Luciani, Jie Luo, Yvonne Lussi, Juan Sebastian Martinez Marin, Pedro Raposo, Daniel L. Rice, James Stephenson, Prabhat Totoo, Nadya Urakova, Preethi Vasudev, Supun Wijerathne, Khawaja Talal Ibrahim, Minjoon Kim, Conny Wing Heng Yu, Alan J. Bridge, Lucila Aimo, Ghislaine Argoud-Puy, Andrea H. Auchincloss, Kristian B. Axelsen, Parit Bansal, Delphine Baratin, Teresa M. Batista Neto, Marie Claude Blatter, Jerven T. Bolleman, Emmanuel Boutet, Lionel Breuza, Cristina Casals-Casas, Kamal Chikh Echioukh, Elisabeth Coudert, Beatrice Cuche, Edouard De Castro, Anne Estreicher, Maria L.

Famiglietti, Marc Feuermann, Elisabeth Gasteiger, Pascale Gaudet, Sebastien Gehant, Vivienne Gerritsen, Arnaud Gos, Nadine Gruaz, Chantal Hulo, Nevila Hyka-Nouspikel, Florence Jungo, Arnaud Kerhornou, Philippe Le Mercier, Damien Lieberherr, Patrick Masson, Anne Morgat, Salvo Paesano, Ivo Pedruzzi, Sandrine Pilbout, Lucille Pourcel, Sylvain Poux, Monica Pozzato, Manuela Pruess, Nicole Redaschi, Catherine Rivoire, Christian J.A. Sigrist, Karin Sonesson, Shyamala Sundaram, Anastasia Sveshnikova, Cathy H. Wu, Chuming Chen, Hongzhan Huang, Kati Laiho, Minna Lehvaslaiho, Peter McGarvey, Darren A. Natale, Karen Ross, C. R. Vinayaka, Yuqi Wang, Raja Mazumber, Vijay Shanker, and Jian Zhang. The UniProt website API: facilitating programmatic access to protein knowledge. *Nucleic Acids Research*, 53(W1):W547–W553, 2025.

25. Alex Bateman, Maria Jesus Martin, Sandra Orchard, Michele Magrane, Aduragbemi Adesina, Shadab Ahmad, Emily H. Bowler-Barnett, Hema Bye-A-Jee, David Carpentier, Paul Denny, Jun Fan, Penelope Garmiri, Leonardo Jose da Costa Gonzales, Abdulrahman Hussein, Alexandr Ignatchenko, Giuseppe Insana, Rizwan Ishtiaq, Vishal Joshi, Dushyanth Jyothi, Swaathi Kandasaamy, Antonia Lock, Aurelien Luciani, Jie Luo, Yvonne Lussi, Juan Sebastian Martinez Marin, Pedro Raposo, Daniel L. Rice, Rafael Santos, Elena Speretta, James Stephenson, Prabhat Totoo, Nidhi Tyagi, Nadya Urakova, Preethi Vasudev, Kate Warner, Supun Wijerathne, Conny Wing Heng Yu, Rossana Zaru, Alan J. Bridge, Lucila Aimo, Ghislaine Argoud-Puy, Andrea H. Auchincloss, Kristian B. Axelsen, Parit Bansal, Delphine Baratin, Teresa M. Batista Neto, Marie Claude Blatter, Jerven T. Bolleman, Emmanuel Boutet, Lionel Breuza, Blanca Cabrera Gil, Cristina Casals-Casas, Kamal Chikh Echioukh, Elisabeth Coudert, Beatrice Cuche, Edouard de Castro, Anne Estreicher, Maria L. Famiglietti, Marc Feuermann, Elisabeth Gasteiger, Pascale Gaudet, Sebastien Gehant, Vivienne Gerritsen, Arnaud Gos, Nadine Gruaz, Chantal Hulo, Nevila Hyka-Nouspikel, Florence Jungo, Arnaud Kerhornou, Philippe Le Mercier, Damien Lieberherr, Patrick Masson, Anne Morgat, Salvo Paesano, Ivo Pedruzzi, Sandrine Pilbout, Lucille Pourcel, Sylvain Poux, Monica Pozzato, Manuela Pruess, Nicole Redaschi, Catherine Rivoire, Christian J.A. Sigrist, Karin Sonesson, Shyamala Sundaram, Anastasia Sveshnikova, Cathy H. Wu, Cecilia N. Arighi, Chuming Chen, Yongxing Chen, Hongzhan Huang, Kati Laiho, Minna Lehvaslaiho, Peter McGarvey, Darren A. Natale, Karen Ross, C. R. Vinayaka, Yuqi Wang, and Jian Zhang. UniProt: the universal protein knowledgebase in 2025. *Nucleic Acids Research*, 53(D1):D609–D617, 2025.
26. Damian Szklarczyk, Katerina Nastou, Mikaela Koutrouli, Rebecca Kirsch, Farrokh Mehryary, Radja Hachilif, Dewei Hu, Matteo E. Peluso, Qingyao Huang, Tao Fang, Nadezhda T. Doncheva, Sampo Pyysalo, Peer Bork, Lars J. Jensen, and Christian Von Mering. The STRING database in 2025: protein networks with directionality of regulation. *Nucleic Acids Research*, 53(D1):D730–D737, 2025.
27. Szymon Chojnacki, Andrew Cowley, Joon Lee, Anna Foix, and Rodrigo Lopez. Programmatic access to bioinformatics tools from embl-ebi update: 2017. *Nucleic Acids Research*, 45(Web Server issue):W550, 2017.
28. Nadezhda T. Doncheva, John H. Morris, Jan Gorodkin, and Lars J. Jensen. Cytoscape StringApp: Network analysis and visualization of proteomics data. *Journal of proteome research*, 18(2):623, 2018.
29. Juexiao Zhou, Haoyang Li, Siyuan Chen, Zhangtianyi Chen, Zhongyi Han, and Xin Gao. Large language models in biomedicine and healthcare. *npj Artificial Intelligence*, 1(1):44, 2025.

30. Xiangbin Meng, Xiangyu Yan, Kuo Zhang, Da Liu, Xiaojuan Cui, Yaodong Yang, Muhan Zhang, Chunxia Cao, Jingjia Wang, Xuliang Wang, Jun Gao, Yuan Geng Shuo Wang, Jia ming Ji, Zifeng Qiu, Muzi Li, Cheng Qian, Tianze Guo, Shuangquan Ma, Zeying Wang, Zexuan Guo, Youlan Lei, Chunli Shao, Wenyao Wang, Haojun Fan, and Yi Da Tang. The application of large language models in medicine: A scoping review. *iScience*, 27(5):109713, 2024.
31. Alexander Verbitsky, Patrick Boutet, and Mohammed Eslami. Metadata harmonization from biological datasets with language models. *Bioinformatics Advances*, 5(1):vbaf241, 2025.
32. Prashanti Manda. Large language models in Bio-Ontology research: A review. *Bioengineering*, 12(11):1260, 2025.